Arun Singh

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Summary

Computer Science graduate (B.Tech, AI & Machine Learning specialization) with a Data Structures & Algorithms foundation in C++ and hands-on experience building, testing, and deploying computer-vision and ML-powered applications end to end. Delivered measurable performance gains – cutting manual processing time by ∼**90%** and improving retrieval precision from 40% to 75% – through systematic testing and benchmarking. Strong grounding in software engineering fundamentals: OOP, REST API development, and full-stack deployment.

Skills

Programming Languages: Python, Java, C, C++, SQL, HTML, CSS, JavaScript

Software Engineering: Data Structures & Algorithms, Object-Oriented Programming (OOP), RESTful APIs, Agile Methodology, Git, GitHub

Backend & Databases: Flask, Streamlit, MySQL, MongoDB, Astra DB, ChromaDB

Machine Learning & AI: Scikit-learn, PyTorch, Hugging Face Transformers, Computer Vision, NLP, Sentiment Analysis, Prompt Engineering, Feature Engineering, Model Training, Model Evaluation, LangChain, Retrieval-Augmented Generation (RAG), OpenAI API

Data Science: Pandas, NumPy, Matplotlib, Seaborn, Statistical Analysis, Data Wrangling, Data Analysis

DevOps & Tools: Docker, Git, Apache Kafka, Grafana, Prometheus, Loki, VS Code, IntelliJ IDEA, Anaconda, Jupyter Notebook

Project

SnapClass - AI Smart Attendance System | GitHub | Live Demo — **December 2025 – January 2026**

- Reduced manual attendance processing time from 15 minutes to 90 seconds per classroom (∼90% reduction), verified by timing 30 sessions pre- and post-automation, by building a real-time pipeline combining SVM-based facial classification (Dlib), voice authentication (Resemblyzer), and QR verification (Segno).
- Delivered real-time analytics dashboards handling concurrent classroom sessions with consistent structured data capture, using a Streamlit application with Pandas-driven aggregation, CSV exports, and Supabase cloud storage.
- Tech Stack: Streamlit, Dlib, Scikit-learn, face_recognition, Librosa, Resemblyzer, NumPy, Pandas, Supabase, bcrypt, Segno.

Art Fusion - Neural Style Transfer Web App | GitHub | Live Demo — **March 2026 – April 2026**

- Implemented AdaIN-based neural style transfer on a pre-trained VGG encoder in PyTorch, engineering configurable alpha-blending for style intensity and validating output quality and inference performance across hardware backends (Apple MPS with CPU fallback).
- Built and deployed a cross-platform image-processing web application – MVC-structured Flask backend with REST API endpoints for image upload and style-blend configuration – achieving stable, reproducible inference on Hugging Face Spaces.
- Tech Stack: Python, PyTorch, Flask, VGG, AdaIN, torchvision, Pillow, Flask-WTF, HTML.

DocQuery AI - PDF Question-Answering System | GitHub — **February 2026 – March 2026**

- Improved retrieval precision from 40% (keyword search) to 75% (RAG) on a 50-query test set spanning legal and technical documents, by building a multi-stage pipeline – PDF ingestion, NLP preprocessing, vector indexing – with LangChain orchestration and Astra DB.
- Achieved sub-2-second query responses over long-form unstructured documents, benchmarked for latency and precision across document types, by applying text segmentation and feature engineering to convert raw PDFs into structured, queryable embeddings.
- Tech Stack: Python, OpenAI API, LangChain, Astra DB, PyPDF2, Streamlit.

Education

Bachelor of Technology in Computer Science — **Graduated: 2026**

Graphic Era Hill University, Dehradun | Specialisation in AI & Machine Learning — *CGPA: 7.85 / 10*

Certifications

Machine Learning Specialization — **Apna Collage – 2026**

Deep Learning Specialization — **Apna Collage – 2026**